



**Course: Database Administration**

**Lecturer: Mr. Thear Sophal**

## **Week 1 – Practical Session**

The review of Entity Relationship Diagram and Relational Model

### **Topics**

1. Blood Donation Management System
2. Mini Mart Online Ordering System
3. Bike Rental Management System
4. Simple Social Media Posting System (Status and Pictures)
5. Music Database System
6. **A Custom Topic of Your Choice (Subject to Approval)**

### **Instructions**

#### **a. Design an Entity Relationship Diagram (ERD)**

Develop an ERD for your selected topic that includes:

- Entities with appropriate notation
- Attributes with appropriate notation
- Relationships with correct notation and cardinality

#### **Submission Format:**

- Initial hand-drawn/sketched version
- Final digital version

#### **b. Create the Database Schema**

Convert your ERD into a **Relational Model (RM)** by:

- Defining tables based on the ERD
- Specifying appropriate data types for each attribute
- Applying constraints, including:
  - Primary Keys
  - Foreign Keys
  - Other relevant constraints
- Establishing relationships between tables

#### **Submission Format:**

- Designed and illustrated using **MySQL Workbench** or **Other tools of your choice**

### c. Implement the Database in MySQL

Create the database and all corresponding tables in **MySQL** using:

- **Structured Query Language (SQL)**
  - Appropriate **Data Definition Language (DDL)** statements

### d. Prepare a Presentation

Develop a presentation slide deck that includes:

- Project Topic Overview
- Entity Relationship Diagram (ERD)
- Relational Model (RM) / Database Schema
- Data Definition Language (DDL) / SQL Code

### Reference Materials (click to see the slide)

- [\*Entity Relationship Diagram\*](#)
- [\*Relational Model\*](#)
- [\*Data Definition Language \(SQL\)\*](#)

### Submission Guidelines

- Submit all project materials through the LMS
- One submission per group only